Computer Networks – Simple Viva Questions & Answers

Basics of Network

1. What is a Network?
   A network is a group of connected computers or devices that communicate and share resources.
2. What is a Link?
   A link is the physical connection between two devices, like a cable or optical fiber.
3. What is a Node?
   A node is any device connected in a network, such as a computer, printer, or router.
4. What is a Router/Gateway?
   A router (gateway) connects two or more networks and forwards data between them.
5. What is Point-to-Point Link?
   A direct connection between exactly two devices.
6. What is Multiple Access?
   When many devices share the same communication medium.

Advantages & Performance

7. Advantages of Distributed Processing
   - Faster problem solving
   - Resource sharing
   - Better security
   - Reliability through backup
   - Collaborative processing
8. Criteria for an Effective Network
   - Performance
   - Reliability
   - Security
9. Factors Affecting Network Performance
   - Number of users
   - Transmission medium
   - Hardware
   - Software
10. Factors Affecting Reliability
    - Frequency of failures
    - Recovery time after failure
11. Factors Affecting Security
    - Unauthorized access
    - Viruses
    - Protocols & Communication
12. What is a Protocol?
    A protocol is a set of rules used for communication between devices.

13. Key Elements of a Protocol
    - Syntax → format of data
    - Semantics → meaning of data
    - Timing → speed and timing of transmission
14. Key Design Issues of a Network
    - Connectivity
    - Resource sharing
    - Common services
    - Performance
15. Define Bandwidth
    Bandwidth is the amount of data transmitted per second.
16. Define Latency
    Latency is the time taken for data to travel from source to destination.
17. What is Routing?
    Routing is the process of finding the path for sending data.
18. What is Peer-to-Peer Process?
    Processes at the same layer communicating with each other.
19. What is Congestion?
    When too many packets arrive and network buffers become full.
20. What is Semantic Gap?
    Difference between application requirements and network capability.
21. What is RTT (Round Trip Time)?
    - Time taken for a message to go and return back.
    - Transmission Types
22. What is Unicasting?
    One sender to one receiver communication.
23. What is Multicasting?
    One sender to selected group of receivers.
24. What is Broadcasting?
    - One sender to all devices in the network.
    - Multiplexing
25. What is Multiplexing?
    Sending multiple signals through a single communication channel.
26. Types of Multiplexing
    FDM
    TDM
    WDM
27. What is FDM?
    Different frequencies are used for different signals.
28. What is TDM?
    Different time slots are given for signals.
29. What is WDM?
    Multiplexing of light signals in fiber optic cables.
30. What is Synchronous TDM?
    Fixed time slots are assigned to each device.

OSI Model

31. Layers of OSI Model
    - Physical Processing
    - Data Link Data
    - Network Need
    - Transport To
    - Session Seems
    - Presentation People
    - Application All
32. Network Support Layers
    - Physical
    - Data Link
    - Network
33. User Support Layers
    - Session
    - Presentation
    - Application
34. Which Layer Connects Both?
    - Transport Layer.
    - Responsibilities of Layers
35. Responsibilities of Physical Layer
    - Bit transmission
    - Data rate
    - Synchronization
    - Topology
36. Responsibilities of Data Link Layer
    - Framing
    - Error control
    - Flow control
    - Physical addressing
37. Responsibilities of Network Layer
    - Logical addressing
    - Routing
38. Responsibilities of Transport Layer
    - Segmentation
    - Flow control
    - Error control
39. Responsibilities of Session Layer
    - Dialog control
    - Synchronization
40. Responsibilities of Presentation Layer
    - Translation
    - Encryption
    - Compression

41. Responsibilities of Application Layer
    - File transfer
    - Email
    - Directory services
    - Transmission Media
42. Hardware Building Blocks of Network
    - Nodes
    - Links
43. Types of Links
    - Cables
    - Wireless links
    - Leased lines
44. Categories of Transmission Media
    - Guided media
    - Unguided media
45. Guided Media Types
    - Twisted pair cable
    - Coaxial cable
    - Fiber optic cable
46. Unguided Media Types
    - Microwave
    - Satellite communication

Error Detection & Correction

47. Types of Errors
    - Single-bit error
    - Burst error
48. What is Error Detection?
    Finding errors in transmitted data.
49. Methods of Error Detection
    - VRC
    - LRC
    - CRC
    - Checksum
50. What is Redundancy?
    Adding extra bits to detect/correct errors.
51. What is VRC?
    Parity bit method for error detection.
52. What is LRC?
    Checks errors using rows of bits.
53. What is CRC?
    Error checking using binary division.
54. What is Checksum?
    Error detection method used in TCP/IP.

55. Steps in Checksum
    - Divide data
    - Add sections
    - Take complement
    - Data Link Concepts
56. What are Data Link Protocols?
    Protocols used for node-to-node communication.
57. Error Detection vs Error Correction
    - Detection → identifies error
    - Correction → identifies and fixes error
58. What is Forward Error Correction?
    Receiver corrects errors using redundant bits.
59. What is Retransmission?
    Resending damaged or lost data.
60. What are Datawords?
    Original data blocks before encoding.
61. What are Codewords?
    Datawords with added redundant bits.
62. What is Linear Block Code?
    XOR of valid codewords gives another valid codeword.
63. What are Cyclic Codes?
    Rotated codeword also remains valid.
64. What is an Encoder?
    Device/software that converts data into coded form.
65. What is a Decoder?
    Device/software that converts coded data back.

Framing

66. What is Framing?
    Dividing data into frames for transmission.
67. What is Fixed Size Framing?
    All frames have the same size.
68. What is Character Stuffing?
    Adding escape characters to distinguish data and flags.
69. What is Bit Stuffing?
    Adding 0 after five consecutive 1s.

Flow & Error Control

70. What is Flow Control?
    Controls speed of data transmission.
71. What is Error Control?
    Detects and corrects transmission errors.
72. What is ARQ?
    Automatic retransmission of lost/damaged frames.
73. What is Stop-and-Wait Protocol?
    Sender sends one frame and waits for acknowledgment.

74. What is Stop-and-Wait ARQ?
    Retransmits frame if acknowledgment is not received.
75. Why Sequence Numbers are Used?
    To identify frames correctly and avoid duplication.
76. What is Pipelining?
    Starting a new task before the previous one finishes.
77. What is Sliding Window?
    Technique to control flow using sequence numbers.
78. What is Piggybacking?
    Sending acknowledgment together with data.

Miscellaneous

79. Types of Transmission Technology
    - Broadcast
    - Point-to-point
80. What is a Subnet?
    A smaller section of a large network.
81. Difference Between Communication and Transmission
    - Transmission → physical sending of data
    - Communication → meaningful exchange of data
82. Modes of Data Exchange
    - Simplex
    - Half duplex
    - Full duplex
83. What is SAP?
    Service Access Point — interface for communication between layers.